V12 Engines — Project Playbook

Version 2 · October 17, 2026. The single source of truth for building the delivery-delay early-warning engine. Replaces the old Team Brief + Project Guidelines + Interface Spec. The dashboard UI lives in the separate Design Handoff. Paste §0 (plus your role's sections) into any AI chat (Claude / Gemini / GPT) before you work.

Passages changed in v2 are shaded blue; the change log at the end lists them.

0. Start here

Project in one line: a real-time, order-time delivery-delay early-warning engine for global health (HIV) supply chains, on IBM LinuxONE. When a shipment is ordered, it predicts late probability + days late, explains the alert, and recommends an action — a logistics officer makes the final call.

Hard constraints (every chat respects these):

- **Honesty:** never fabricate numbers or causes; flag logical/data errors first. Historical data is demoed as a replay, never claimed as a live 2026 feed.
- **Build timing (rules):** all project code is written DURING the event (Oct 17–18). Before that: planning, design, practice, skeletons only — no pre-built submission.
- **Platform:** IBM LinuxONE, s390x, NO GPU. Models = Snap ML boosted trees + logistic baseline; generative = IBM Granite. Keep models simple (tabular).

Which sections to read:

- Data / preprocessing: §1, §3, §4. Model: §3, §4. API & genAI: §4, §5. Platform: §4, §6. Dashboard: §0, §4 + the Design Handoff.

Working agreement: everyone codes to the §4 contract (fixed shapes); one owner per area + backup; small PRs (task→area→main); the demo VM runs main only; if a chat conflicts with this doc, the doc wins.

1. Data facts (verified Oct 16 from the SCMS CSV)

- **Dataset:** USAID/PEPFAR shipments — 10,324 rows, 2006–2015, 43 countries. HIV treatment drugs (ARVs) & test kits — NOT vaccines. Mix: ARV 8,550 · HRDT 1,728 · small malaria set 46.
- **Labels clean:** scheduled & delivered dates 100% present → late computed exactly. Overall late rate 11.5% (no-skill baseline 0.115).
- **Delays concentrate by fulfillment route:** Direct Drop 5.3% vs From RDC 17.2%; From RDC = 78% of all late shipments. Keep From RDC — it is the target.
- **Shipment mode:** Air 6,113 · Truck 2,830 · Air Charter 650 · Ocean 371 · blank 360. Blanks (1.1% late) are NOT South-Africa-specific and are NOT Ocean (17.5% late) → keep as "Unknown".
- **Order-time anchor partly missing by design:** PO date absent 55.5% (From RDC has none). Decision date = PO else PQ covers 81.3%; planned lead time median 128 days.
- **Missing anchors are concentrated in 2006–2009** (1,931 of 7,779 training rows; 86% From RDC; only 0.8% late vs 13.9% for rows with an anchor). Every test row (2014–2015) has one.
- **Non-stationary:** late rate swings by year (e.g. From RDC 5%→37%→10%→35%) — motivates recency windows and "retrain on current data."
- **Honest limits:** no dispatch date / in-transit location / temperature → no transit-time or cold-chain modeling. Historical → deployment retrains on a current feed.

2. Pitch & framing

- **Hook (Uber):** ride apps predict arrival to the minute; life-critical health deliveries still have no order-time prediction.
- **Honest problem:** officers DO sense delays via calls/emails with carriers & forwarders, but that signal is manual and inconsistent, and global logistics is disrupted by the unexpected (customs delays, cancelled flights, port congestion) → signals get missed or come too late. Never say "they only find out after arrival."
- **Urgency (honest):** stakes = HIV treatment continuity (late ARVs/test kits → interrupted treatment — not "vaccine"). Malaria = tiny real subset + roadmap example.
- **Video line:** "We replay real 2006–2015 PEPFAR shipments as if live to demo scoring. The model trains on earlier shipments, is evaluated on later ones, and in deployment retrains on current data."

3. Modeling decisions (confirmed)

- **Label:** late = delivered > scheduled (11.5%). Order/decision date is NOT part of the label (only split + runway). Second model = days-late regressor.
- Keep From RDC; shipment-mode blanks → "Unknown" (never impute to Ocean). weight EXCLUDED (not known at order time).
- **Decision date** = PO else PQ (81.3%). When neither exists it stays **empty** with decision_date_known = 0. It is never filled with another date (it is the date the order was placed).
- **Prior-rate cutoff (prior_cutoff_date)** = the decision date, else an **estimated order date**: scheduled − median planned lead time for the shipment's fulfillment route (Direct Drop 91 d, From RDC 162 d), learned from training-period rows with a real anchor only. Capped at the scheduled date; flagged cutoff_estimated = 1. Used only to cut off history — not a feature.
- **planned_lead_time** = scheduled − decision_date (NaN for ~19%, plus 4 rows with a negative lead; kept with the flag, never dropped).
- **Split (Option B):** by scheduled date, chronological — train ≤2013 (7,779 rows, 830 late) / test 2014–2015 (2,545 rows, 356 late). ~25% test, enough late cases for stable PR-AUC/recall. Tune thresholds on a 2013 validation slice of the training data, never on test (test late rate 14.0% vs train 10.7%).

Error framing — what we optimize

- **Goal = minimize errors, not maximize accuracy.** Type II (predicted on-time, actually late → missed warning → treatment interruption) is costliest → favor recall. Type I (false alarm) → alert fatigue. Report Correct / Type I / Type II as a share of the test set.
- **Imbalance 11.5%** → class weights / threshold tuning; report PR-AUC & recall at the operating threshold, not accuracy. Surface only the top ~10% by risk (alert-fatigue control).

Final feature set (16) — matches the schema in §4

- country, vendor, fulfillment_route, shipment_mode, product_group, managed_by,
- manufacturing_site, quantity, planned_lead_time_days, decision_date_known,
- prior_late_rate_{country,vendor,fulfillment_route}, prior_known_{country,vendor,fulfillment_route}
- **Note on decision_date_known:** it is 1 for every test row and every live order, so it can only learn the 2006–2009 gap. The model owner reports three variants — with it, without it, and trained on

2010–2013 only — and the team keeps whichever scores best on the 2013 validation slice. If that drops the feature, update this list to 15.

prior_late_rate design (confirmed)

- For each shipment, the prior late-rate of a key uses ONLY shipments DELIVERED before this row's cutoff (**cutoff = prior_cutoff_date**: decision date, else the estimated order date), within a 365-day window, smoothed toward the same-window global prior with M = 20 pseudo-counts. **The shipment itself is excluded from both its key's count and the global prior.** Leakage-safe by construction. Cold start / no prior → fallback 0.115 and prior_known=0.
- Lane = each of country / vendor / route separately (no pairs) for now; pairs are a later enhancement. Verified: From RDC prior ≈ 0.137, and late vs on-time are separated (country prior 0.14 vs 0.09).

Models

- **Baseline:** class-weighted Logistic Regression. **Main:** Snap ML boosted-tree classifier (late probability) + regressor (days late). **Generative explanations:** IBM Granite. Must beat baselines on the test split.

4. Interface & Contract (the glue between workstreams)

Treat field names and JSON shapes as fixed unless the team changes them here.

Repo & artifacts

- Branches: task → area → main (PRs). Folders src/features/ src/training/ src/api/ src/replay/ src/genai/ src/web/ docs/. VM paths /srv/v12e/{data,models,state}.

Cleaned data schema (output of preprocessing, Parquet)

```
order_id(str) country(cat) vendor(cat) fulfillment_route(cat: Direct Drop|From RDC)
shipment_mode(cat: Air|Truck|Air Charter|Ocean|Unknown) product_group(cat)
managed_by(cat) manufacturing_site(cat) quantity(num)
decision_date(date|null) decision_date_known(0/1) planned_lead_time_days(num|null)
scheduled_delivery_date(date) prior_late_rate_{country,vendor,fulfillment_route}(num)
prior_known_{country,vendor,fulfillment_route}(0/1)
prior_cutoff_date(date) cutoff_estimated(0/1)          # preprocessing only, NOT features
label_late(0/1) days_late(int>=0)                      # targets
```

Model artifact contract

- Two models in /srv/v12e/models: classifier (late prob) + regressor (days late), Snap ML native or ONNX, same feature order as above, carry a model_version; identical predictions after save→reload.
- **Missing planned_lead_time_days** is handled inside build_features so training and scoring match: the logistic baseline always gets the **training** median (stored with the artifacts, never recomputed from requests); the Snap ML models get the same fill unless Snap ML on s390x is confirmed to accept missing values. decision_date_known acts as the missing flag.

Scoring API — POST /score (request → response)

```
REQ  {order_id, country, vendor, fulfillment_route, shipment_mode, product_group,
      managed_by, manufacturing_site, quantity, decision_date, scheduled_delivery_date}
RESP {order_id, model_version, late_probability:0..1, predicted_days_late:int,
      runway_days:int, runway_tier:'long|medium|short', risk_band:'high|medium|low',
      top_drivers:[{feature,value,direction}], explanation:str, recommended_action:str}
```

- **decision_date is required** (for a live order it is the moment the order is placed). If it is missing or after scheduled_delivery_date, return {"error":{"code":"invalid_decision_date","message":...}} with HTTP 422 — never guess it.
- GET /alerts scored list, ranked risk × quantity × product-criticality, shorter runway first among ties.
- GET /stream SSE replay feed (one scored shipment per event). The replay uses test rows only, which all have a real decision date. GET /metrics → {late_rate, no_skill:0.115, pr_auc, recall_at_threshold, confusion:{tp,fp,fn,tn}, error_breakdown:{correct,type1,type2}, split:'train<=2013 / test 2014-2015'}.
- **Errors:** {"error":{"code","message"}} + proper HTTP status. Validate inputs; never crash on a bad field.

Generative layer (Granite) — structured output

```
{risk_summary:str, recommended_action:str (matched to runway_tier), caveat:str}
```

- **Guardrails:** ground every statement in the provided top_drivers — never invent causes; never claim certainty; predicted days late is an estimate; action must match the runway tier; if an input such as lead time is missing, say it is unknown.

Shared enums & thresholds

- runway_tier: long(>120d) | medium(45–120d) | short(<45d) · status: New→Acknowledged→Actioned→Dismissed · risk_band by late_probability percentile (focus top ~10%).

Dashboard data contract

- Live ← /stream (alert cards: prob, days late, runway badge, drivers, explanation, action, 3 HITL buttons). Model ← /metrics (single time-split scorecard + Error breakdown; NO weekly slider). Platform = environment evidence. Full UI spec in the Design Handoff.

5. Security by design (framing)

- Scored dimension and the core of "Why IBM Z." We USE the platform's security, not build our own: TLS on the scoring API, isolated container, IBM Z pervasive encryption. Mark only what you enable "Verified"; quantum-safe = roadmap.
- **Data governance:** de-identified (shipment-level, no patient data); access control; the human-in-the-loop log is the audit trail.

6. Build & ops

- **Event code = SSF2026** (confirmed Oct 16; not SSF2025). Required at L1CC registration for the AI-Lab server; also in the project description.
- **One server per TEAM.** Choose a "-datathon" image (Ubuntu 24.04-datathon / RHEL 9.8-datathon) + the "AI Lab" flavor. Click the activation email; set your own Jupyter token (public IP).
- **Reference lab:** IBM's "Predictive + Generative (loan approval)" lab = our hybrid structure; use as code template.
- **Submission (Hack2Skill):** max 3-min demo video (problem→data→solution→LinuxONE demo) + public GitHub/GitLab link + title [V12 Engines_IBM Z Datathon 2026 Project] + open-source links. Flowchart & working prototype = recommended optional.

Change log

v2 · Oct 17, 2026 (decided at the event; preprocessing notebook v2 implements and tests it)

- §1: added where missing decision dates are concentrated (2006–2009; none in test).
- §3: decision date is never filled; new prior_cutoff_date (estimated order date per route, training-period medians, capped at scheduled, flagged cutoff_estimated); self excluded from global prior too; thresholds tuned on a 2013 validation slice; three-variant check for decision_date_known.
- §4: schema — decision_date nullable, prior_cutoff_date + cutoff_estimated added (not features); artifact contract — training-median fill for missing lead time inside build_features; /score — decision_date required, 422 if missing or after scheduled; replay uses test rows only; Granite states unknown inputs as unknown.
- Effect on data: train priors changed on 1,847 rows (mean change ≈0.01 on rows without a decision date); test priors changed on 4 rows; labels, dates, lead times and split counts unchanged.